

Operational Definition of a Target Pump

1. General definition

A **target pump** is a sustained transition of a previously dormant, low-activity cryptocurrency into a new market regime characterized by:

- an abrupt and abnormally large increase in trading volume;
- a visible expansion of price movement relative to the preceding dormant period;
- predominantly upward and persistent price movement;
- continued elevated trading activity;
- limited pullbacks during the expansion phase;
- movement that is substantially stronger than the simultaneous movement of Bitcoin or the broader market.

The event must originate from a previously dormant market state. A strong move in an already actively traded or highly volatile asset is not considered the target setup.

2. Dormant baseline state

Before the pump begins, the asset must demonstrate a sufficiently long period of low market activity.

Recommended baseline window:

7–30 days before the event

The exact duration should be configurable.

A market is considered dormant when most of the following conditions are present:

1. Low median quote volume on five-minute candles.
2. Low number of trades or low trading frequency.
3. Small median five-minute price range.
4. Low realized volatility.
5. Long periods during which the chart visually resembles a narrow line.
6. No persistent directional trend.
7. Weak or inconsistent short-term relationship with Bitcoin.
8. No comparable volume or volatility expansion shortly before the event.

Dormancy should be measured relative to the asset's own previous behavior rather than using one universal absolute-volume threshold.

Suggested baseline values:

```
baselineVolume5m =  
    median five-minute quote volume during the baseline window  
  
baselineRange5m =  
    median five-minute high-low range during the baseline window  
  
baselineVolatility =  
    median or EWMA volatility during the baseline window
```

Extremely low liquidity does not automatically make an event invalid. Low liquidity is part of the target setup.

However, a separate flag may indicate that the market was too illiquid to trade safely:

```
untradeableLiquidity = true | false
```

3. Pump start

The pump start time, `t0`, is the first five-minute candle at which both market activity and price behavior clearly leave the preceding dormant regime.

The start should require both:

1. Abnormal trading activity.
2. Upward price expansion.

A volume spike without a meaningful price response is not sufficient.

Suggested start conditions:

```
volumeRatio5m =  
    currentVolume5m / baselineVolume5m  
  
rangeRatio5m =  
    currentRange5m / baselineRange5m
```

An initial candidate may be created when:

```
volumeRatio5m >= configuredVolumeThreshold  
AND  
rangeRatio5m >= configuredRangeThreshold
```

```
AND  
return5m > 0
```

Reasonable initial values for experimentation:

```
configuredVolumeThreshold: 8-15  
configuredRangeThreshold: 3-5
```

These numbers are starting parameters, not the final definition. They must later be evaluated against the manually labeled dataset.

The detected start should be the earliest candle satisfying the conditions, not the candle on which the move becomes visually obvious after the fact.

4. Early pump candidate

The screener's objective is to identify a likely pump during its first three to five five-minute candles.

Target detection window:

```
15-25 minutes after t0
```

At detection time, only data that already exists may be used.

An early candidate should demonstrate the following properties.

4.1 Persistent volume

The event should not consist of only one abnormal candle.

Volume should remain substantially above the dormant baseline during multiple consecutive candles.

Example:

```
At least 2 of the first 3 candles:  
volume5m >= 5 × baselineVolume5m
```

4.2 Directional price movement

The cumulative price return should be positive and significant relative to the asset's dormant volatility.

Example:

```
return15m >= max(  
    configuredMinimumReturn,  
    configuredVolatilityMultiplier × baselineVolatility  
)
```

A starting experimental threshold may be:

```
return15m >= 5%
```

but the final value should be derived from the labeled history.

4.3 Movement efficiency

The price should move predominantly upward rather than oscillating violently in both directions.

A useful directional-efficiency metric is:

```
directionalEfficiency =  
    net upward movement /  
    total absolute movement
```

Conceptually:

```
directionalEfficiency =  
    (lastClose - initialOpen) /  
    sum(abs(close[i] - close[i - 1]))
```

Interpretation:

```
close to 1.0 = almost uninterrupted upward movement  
close to 0.0 = substantial noise and two-sided movement
```

A target pump should normally have relatively high directional efficiency during its early phase.

4.4 Limited pullback

Calculate the maximum decline from the running high during the detection period:

```
earlyMaxDrawdown =  
    maximum decline from a previous local high  
    between t0 and detection time
```

A pump candidate should not repeatedly lose most of its upward progress.

The initial threshold may be approximately:

```
earlyMaxDrawdown <= 3-5%
```

However, this threshold describes the desired trading setup rather than the universal definition of a pump.

4.5 Market independence

The move should be substantially stronger than the simultaneous market movement.

A simple first version:

```
abnormalReturn15m =  
    coinReturn15m - bitcoinReturn15m
```

A candidate should require:

```
abnormalReturn15m >= configuredMarketExcessReturn
```

For example, a coin rising by 7% while Bitcoin rises by 0.5% is materially different from a coin rising by 7% while the entire altcoin market rises by 6%.

A more advanced beta-adjusted calculation may be added later, but it is not required for the initial implementation.

5. Confirmed target pump

An event is retrospectively labeled as a **confirmed target pump** when the initial regime change develops into sustained upward activity.

The confirmation window begins at `t0`.

Recommended minimum duration:

60 minutes

A confirmed target pump should satisfy most or all of the following conditions:

1. Trading volume remains materially above the dormant baseline for a substantial part of the following hour.
2. Price continues to establish higher highs or materially higher trading levels.
3. The movement is not explained by a simultaneous broad-market rally.
4. The event is not limited to a single wick or isolated trade.
5. Price does not immediately return to the pre-event range.
6. The cumulative move is materially larger than ordinary volatility.
7. The pump remains visually and numerically distinguishable from the preceding dormant period.

Suggested measurable confirmation conditions:

```
duration >= 60 minutes

AND

maximumReturn60m >= configuredPumpReturn

AND

numberOfElevatedVolumeCandles >= configuredMinimumCount

AND

directionalEfficiency60m >= configuredMinimumEfficiency
```

Possible initial experimental values:

```
maximumReturn60m >= 10-15%

at least 6-8 of 12 five-minute candles
have volume >= 3 × baselineVolume5m
```

These are provisional values to test against the manually classified events.

The human label remains the source of truth during the first iteration.

6. End of the pump

The pump does not end merely because one red candle appears.

The end time, `tEnd`, is the earliest point at which the elevated directional regime has clearly broken.

The event may be considered finished when at least two of the following conditions occur:

1. No new price high is established for three consecutive five-minute candles.
2. Price falls materially from the running pump high.
3. Volume returns close to the dormant baseline.
4. Price returns to the pre-pump range.
5. The recent fifteen-minute net return becomes negative.
6. The market enters a clear distribution or dump phase.

Suggested initial rules:

```
no new high for at least 15 minutes
```

```
AND/OR
```

```
drawdown from pump high >= 7-10%
```

```
AND/OR
```

```
volume5m < 2 × baselineVolume5m  
for 3 consecutive candles
```

The exact end-of-pump threshold should not affect whether the event was correctly detected near its beginning.

7. Target pump versus tradeable pump

A real pump and a pump that is suitable for a specific trading strategy are not the same label.

The primary label should answer:

```
Did a sustained target pump occur?
```

A separate strategy-specific property should answer:

Could the event be traded using the selected entry and a 5% price stop?

Recommended derived fields:

```
interface PumpTradeability {  
  entryTimestamp: string;  
  entryPrice: number;  
  
  maxAdverseExcursionBeforeTarget: number;  
  maxFavorableExcursion: number;  
  
  reachedProfit10Percent: boolean;  
  reachedProfit15Percent: boolean;  
  reachedProfit20Percent: boolean;  
  
  stoppedAt5PercentBeforeTarget: boolean;  
}
```

This separation prevents a real pump from being labeled as a false positive merely because the chosen entry was late or the initial pullback exceeded the strategy's stop.

8. Recommended primary categories

The six primary annotation categories should be adjusted to match the target definition.

1. `target_pump`

A sustained dormant-asset pump satisfying the intended setup.

Typical characteristics:

- dormant baseline;
- abrupt volume regime change;
- sustained directional growth;
- elevated activity for approximately one hour or longer.

2. `short_directional_move`

A genuine directional move occurred, but it did not persist long enough to qualify as the target pump.

Examples:

- meaningful growth lasting 15–45 minutes;
- several strong candles followed by the end of activity;
- a real impulse, but not the desired sustained setup.

This class is important because it is not the same as a wick spike.

3. `wick_spike`

A short-lived price excursion with immediate or near-immediate retracement.

Typical characteristics:

- one or a few candles;
- large upper wick;
- no sustained volume;
- no continuation.

4. `volume_only`

A substantial activity or volume increase occurred without a meaningful directional price move.

Typical characteristics:

- high volume;
- many trades;
- price remains inside a range;
- strong absorption or two-sided trading.

5. `market_move`

The price moved because the broader market or sector moved at the same time.

The event did not demonstrate sufficient independent abnormal performance.

6. `unclear`

The event cannot be reliably classified because of ambiguity, missing data, poor chart quality, or conflicting characteristics.

9. Secondary flags

Conditions that do not represent separate event types should be stored as flags.

```
interface PumpAnnotationFlags {
    untradeableLiquidity?: boolean;
    badMarketData?: boolean;
    detectionTooLate?: boolean;
    likelyWashTrading?: boolean;
    possibleNewsDrivenMove?: boolean;
    broadMarketContribution?: boolean;
}
```

In particular, `illiquid_noise` should not remain a primary category because baseline illiquidity is an expected property of the target assets.

Instead:

```
Primary category: wick_spike
Flag: untradeableLiquidity
```

or:

```
Primary category: unclear
Flag: badMarketData
```

10. Compact formal rule

The target event can be summarized as:

```
A target pump is a transition from a prolonged dormant market regime
to an abnormal and persistent regime of elevated trading volume
and predominantly upward price expansion, where the movement is
substantially stronger than the broader market and remains active
for approximately one hour or longer.
```

The early screener should estimate:

```
P(
    confirmed target pump during the next 60 minutes
    |
    first 15-25 minutes after the detected regime change
)
```

The retrospective human annotation answers whether the complete event actually became a confirmed target pump.